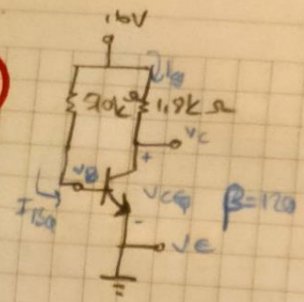
Buğra YALÇIN 202313709026 Elektronik Devreler Ödev 4

Transistör Soruları- 1 17/12/2024

1)



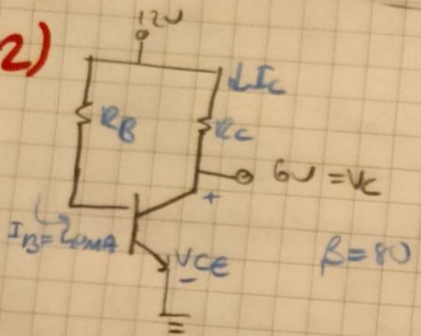
a) $I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{16-0,7}{510k} = \underline{30\mu A}$

b) $I_{CQ} = 120 \cdot 30\mu A = \underline{3,6mA}$

c) $V_{CEQ} = V_{CC} - I_C \cdot R_C = 16 - 3,6mA \cdot 1,8k = \underline{9,52V}$

d) $V_C = \underline{9,52V}$ e) $V_B = \underline{0,7V}$ f) $V_E = \underline{0V}$

2)



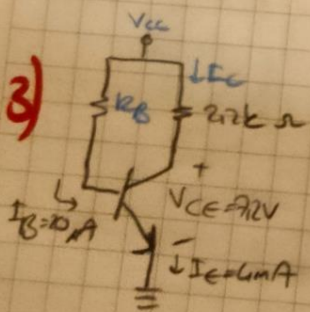
a) $I_C = I_B \cdot \beta \Rightarrow 80 \cdot 40\mu A = \underline{3,2mA}$

b) $R_C = \frac{V_{CC} - V_C}{I_C} = \frac{12-6}{3,2mA} = \underline{1,875k\Omega}$

c) $R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{12-0,7}{40\mu A} = \underline{282,5k\Omega}$

d) $\underline{V_{CE} = 6V}$

3)



a) $I_C = I_E - I_B \Rightarrow 4mA - 20\mu A = \underline{3,98mA}$

b) $V_{CC} = I_C \cdot R_C + V_{CE} = 3,98mA \cdot 2,2k + 7,2$

$= \underline{15,956V}$

c) $\beta = \frac{I_C}{I_B} = \frac{3,98mA}{20\mu A} = \underline{199}$

d $= R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{15,956 - 0,7}{20\mu A} = \underline{762,8k\Omega}$

(1

4) soru1'deki $I_{C_{sat}} = \frac{V_{CC}}{R_C}$ $\frac{16}{1,8k} = 8,88mA$

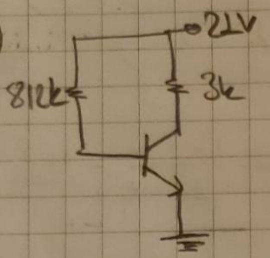
5) a) $I_C = \frac{V_{CC}}{R_C} = \frac{21}{3k} = 7mA$ b) $= R_B = \frac{V_{CC} - V_{BE}}{I_B} = \frac{21-0,7}{25\mu A} = 812k\Omega$

c) $I_{CQ} = 3,5mA$ $V_{CEQ} = 10,5V$ d) $\beta = \frac{I_C}{I_B} = \frac{3,5mA}{25\mu A} = 140$

e) $\alpha = \frac{\beta}{\beta+1} = \frac{140}{141} = 0,993$ f) $I_C = \frac{V_{CC}}{R_C} = \frac{21}{3k} = 7mA$

(5

g)



h) $P_d = V_{CE} \cdot I_C = 10,5 \cdot 3,5mA = 36,75mW$

i) $P_S = V_{CC} \cdot (I_B + I_C) = 21 \cdot (25\mu A + 3,5mA) = 74mW$

j) $P_d - P_S = 74 - 36,75 = 37.25mW$

6) a) $V_{CE} = V_{CC} - I_C . R_C \Rightarrow I_C = 8,88mA$

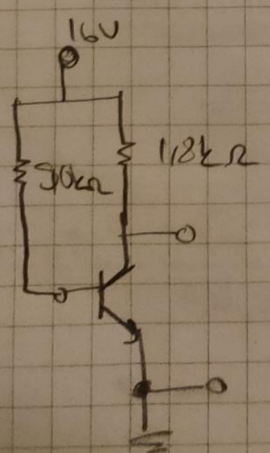
$0 = 16V - I_C . 1,8k$

b) $= Q\left(\frac{V_{CE}}{2}, \frac{I_C}{2}\right) = Q(8V, 4,45)$

(3

c) $\beta = \frac{I_C}{I_B}$ $\beta = \frac{4,45}{30\mu A} = 148$

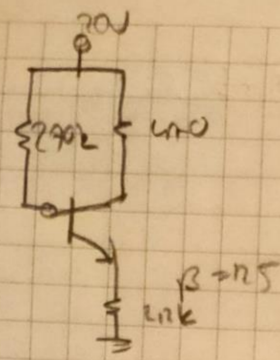
7)



$I_C = \beta . I_B = 120,148\mu A = 2,102mA$

$V_{CE} = 16 - 2,02 . 1,8 = 12,36V$

8)



a) $I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta+1) R_E} = \frac{20 - 0,7}{240k + 126 \cdot 2,2k} = 35 \mu A$

b) $I_C = \beta \cdot I_B = 125 \cdot 35 \mu A = 4,375 mA$

c) $V_{CE} = V_{CC} - I_C (R_C - R_E) \Rightarrow 20 - 4,375 \cdot (\ldots)$

$= 8,132V$

d) $= V_C = 20 - 4,375 \cdot 0,430$

$= 17,84V$

e) $V_B = 20 - 2,7435$

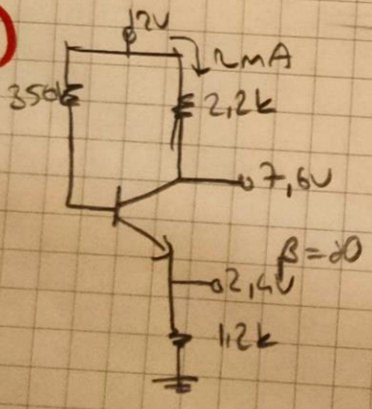
$= 10,55V$

f) $V_E = 9,62$

9) a) $I_C = \beta \cdot I_B = 125 \cdot 28 = 3,5 mA$ b) $Q(10V, 3,75mA)$

c) $\beta = \frac{3,75 mA}{28 \mu A} = 133$ d) $= 133 > 125$

10)



a) $R_C = \frac{V_{CC} - V_C}{I_C} \Rightarrow \frac{12 - 7,6}{2mA} = 2,2k\Omega$

b) $R_E = \frac{2,4 - 0}{2mA} = 1,2k\Omega$

c) $I_C = \beta \cdot I_B \Rightarrow 2 = 80 \cdot I_B \Rightarrow I_B = 25 \mu A$

d) $V_{CE} = 5,2V$ e) $V_B = 3,1V$

11) a) $\beta = \frac{3{,}09}{20\mu A} = 154$ b) $V_{CC} = V_C + I_C . R_C = 17{,}74$ (✓

c) $R_B = \frac{17{,}74 - 2{,}8}{20\mu A} = 747k\Omega$

12) $I_C = \frac{V_{CC} - V_{EE}}{R_C + R_E} = \frac{20-0}{9{,}4 + 17k} = 7{,}49\,mA$

13) a) $I_C = \frac{V_{CC} - V_{CE}}{R_C + R_E} = \frac{4 = 24 - 10}{12k + R_E} = R_C = 2{,}3k$

(c

b) $\beta = \frac{4mA}{30\mu A} = 133$ c) $= I_B = \frac{V_{CC} - V_{BE}}{R_B + R_{E}.\beta} \Rightarrow 30\mu A = \frac{24 - 0{,}7}{R_B + 1{,}2k . 133}$

$R_B = 617k$

(a